

Report
Of
PYTHON FULL STACK WEB DEVELOPMENT

Submitted To
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In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, KRITIKA AGARWAL, do hereby declare that the internship report titled "PYTHON FULL STACK WEB DEVELOPMENT" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement, and EduSkills Academy, under the patronage of AICTE, for providing me the opportunity to undertake this virtual internship and gain valuable industry-relevant skills.

I express my gratitude to my parents for their blessings.

Signature: _____

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TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate	iii
4.	Project Description	
	4.1 Introduction	1
	4.2 Organization Profile	
	4.3 Problem Statement	
	4.4 Project Objectives	
	4.5 Scope of the Project	
	4.6 Technologies and Tools Used	
	4.7 System Architecture	
	4.8 Methodology	
	4.9 Expected Outcomes	
	4.10 Certificate of Completion & Communication Proof	
5.	Bibliography/References	

Note: Page numbers to be finalized and updated after final printing/pagination, per the prescribed format instructions.

3. INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by EduSkills Academy, under the patronage of AICTE (National Internship Portal), certifying successful completion of the 8-Week AICTE – EduSkills Virtual Internship Program in Python Full Stack Development, is to be attached on this page as a scanned copy/image.

[ATTACH SCANNED COPY OF THE INTERNSHIP COMPLETION CERTIFICATE HERE]

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during the 8-Week AICTE – EduSkills Virtual Internship Program in the domain of Python Full Stack Development, undertaken from June 2026 – August 2026 (8 Weeks). The internship was designed to provide structured learning combined with practical exposure to modern web development technologies, covering the complete stack from front-end design to back-end development and database management.

The program followed a week-wise structured learning plan comprising weekly assessments, project documentation submissions, and a final assessment test, culminating in the award of a Virtual Internship Certificate upon successful completion of all requirements.

4.2 Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO 27001:2013 certified, Great Place to Work Certified organization operating under the mission "Nation Building Through Skills." EduSkills partners with AICTE (All India Council for Technical Education) under the National Internship Portal to design and deliver industry-aligned virtual internship programs for engineering and technology students across India.

These programs are structured to bridge the gap between academic learning and industry requirements by offering hands-on, skill-based training in emerging technology domains, supported by weekly assessments and a capstone project-based evaluation.

4.3 Problem Statement

Modern web applications require developers who are proficient across the entire technology stack — from designing responsive, accessible user interfaces to building robust server-side logic and managing persistent data efficiently. Many students, however, gain exposure to only isolated parts of this stack during their academic coursework.

The problem addressed by this internship was to acquire and demonstrate end-to-end full stack development competency: building a complete, functional web application using front-end technologies (HTML, CSS, Bootstrap, JavaScript, jQuery) integrated

with a back-end framework (Django) and a relational database (MySQL), following industry-standard development practices.

4.4 Project Objectives

- To understand and apply the fundamentals of HTML and CSS for structuring and styling web pages.
- To build responsive, mobile-first web interfaces using the Bootstrap framework.
- To implement client-side interactivity and DOM manipulation using JavaScript and jQuery.
- To learn Python programming fundamentals, including OOP concepts, for back-end development.
- To design and develop a dynamic, data-driven web application using the Django framework.
- To design a relational database schema and perform CRUD operations using MySQL, integrated with Django.
- To gain practical, hands-on experience through weekly assessments and a project-based evaluation.

4.5 Scope of the Project

The scope of this internship encompassed the design and development of a full stack web application, covering the following areas:

- Front-end design and layout using HTML5, CSS3, and Bootstrap for responsiveness.
- Client-side scripting and interactivity using JavaScript and jQuery.
- Back-end application logic, routing, and templating using the Django framework.
- Database design, schema creation, and CRUD operations using MySQL.
- User authentication and session management within the Django application.
- Integration and testing of the complete application stack from front end to database.

The scope was limited to the learning modules and capstone assignment prescribed under the 8-week virtual internship curriculum and did not extend to production deployment or third-party payment/API integrations.

4.6 Technologies and Tools Used

- Front-End: HTML5, CSS3, Bootstrap 5, JavaScript, jQuery
- Back-End: Python 3, Django Framework
- Database: MySQL
- Development Tools: Visual Studio Code, Git & GitHub, Web Browser Developer Tools
- Others: Postman (API testing), XAMPP/MySQL Workbench (database management)

4.7 System Architecture

The application follows a three-tier architecture consisting of a presentation layer, an application (logic) layer, and a data layer:

- Presentation Layer: Built using HTML, CSS, Bootstrap, JavaScript, and jQuery, this layer is responsible for rendering the user interface and handling client-side interactions.
- Application Layer: Implemented using the Django framework (Python), this layer handles request routing, business logic, form processing, authentication, and template rendering (Model-View-Template pattern).
- Data Layer: MySQL serves as the relational database management system, storing and managing persistent application data, accessed through Django's Object-Relational Mapping (ORM).

Requests from the client browser are routed through Django's URL dispatcher to the appropriate view, which interacts with the MySQL database via Django models and returns a rendered HTML template as the response, styled using Bootstrap and enhanced with JavaScript/jQuery for dynamic behaviour.

4.8 Methodology

The internship followed a structured, week-wise learning methodology, combining self-paced module study, weekly assessments, and hands-on project work, as summarised in Table 4.1 below:

Table 4.1: Week-wise Learning Methodology

Week	Module	Description
Week 1	HTML	Page structure, elements, forms, tables, semantic tags, and website layout fundamentals.
Week 2	CSS	Selectors, box model, positioning, flexbox, grid, responsiveness, and styling techniques.
Week 3	Bootstrap	Responsive, mobile-first design using Bootstrap components, grid system, and utilities.
Week 4	JavaScript	Variables, functions, loops, conditionals, DOM manipulation, and event handling.
Week 5	jQuery	DOM traversal, animations, event handling, and AJAX interactions.
Week 6	Python Programming	Syntax, data types, loops, functions, OOP concepts, and backend fundamentals.
Week 7	Django Framework	Models, views, templates, routing, and authentication systems.
Week 8	MySQL & Database Management	SQL fundamentals, database design, CRUD operations, and MySQL–Django integration.

Each module was followed by a weekly assessment to evaluate conceptual understanding, and select modules required submission of project documentation and assignments. The final week concluded with a comprehensive Final Assessment Test, and the overall grade was computed based on weekly assessments, project submissions, and the final test.

4.9 Expected Outcomes

- Practical proficiency in building responsive, well-structured web pages using HTML, CSS, and Bootstrap.
- Ability to add client-side interactivity using JavaScript and jQuery.
- Working knowledge of Python programming and object-oriented concepts.
- Hands-on experience developing and deploying a dynamic web application using Django.
- Competency in relational database design and CRUD operations using MySQL.

- A complete, functioning full stack web application demonstrating integration of all the above technologies.
- Enhanced employability and industry readiness in the field of full stack web development.

4.10 Certificate of Completion and Other Communication Mail Proof

The Internship Offer Letter (Ref. No. C17Y2026M081516899) issued by EduSkills, and the Internship Completion Certificate (to be issued upon successful completion of all weekly assessments and the Final Assessment Test), along with any related communication/email proof, are attached in Chapter 3 and in the Appendix of this report.

5. BIBLIOGRAPHY/REFERENCES

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